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Formation Challenges of Lithium-Ion Battery Manufacturing

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Introduction

This paper discusses the critical importance of reducing the electrolyte wetting, formation, and aging times associated with lithium-ion battery (LIB) manufacturing. These steps are essential for ensuring high-quality LIBs with uniform capacity, safety, and long cycle life, but they add great expense to the manufacturing cost, as wetting and formation may take 3–7 days and aging may take up to an additional 2 weeks. These steps account for a considerable portion of production plant capital expenses and can take up to 25% of the floor space. Wetting and formation are also necessary for building the anode solid electrolyte interface (SEI) and cathode electrolyte interface (CEI). Aging is necessary for determining whether leakage currents are too high before the rated capacity is assigned to a LIB. The SEI and CEI formation process, the effect of formation protocols themselves, and the importance of electrolyte imbibition into the electrode and electrolyte pores are considered in the context of the overall industrial challenges associated with these three "end of cell production" steps. The original data presented here are discussed in a broader context of

lowering the production and capital costs of manufacturing LIBs, and an outlook is given on the implementation of these technologies by U.S. battery manufacturers. Other technologies, such as addition of surface coatings of the active material powders, are considered as possible solutions to further reduce or even eliminate wetting, formation, and aging altogether.

Important advances in LIB active materials, electrode design, energy density, and cell design have recently been implemented,[1] but key manufacturing challenges remain in order to lower cell costs for widespread transportation and grid storage commercialization.[2] The anode SEI and CEI formation step is one of the most critical aspects of the production of LIBs for a variety of reasons. First, the delicate active material interface layers need to form in a controlled manner through reactions with the electrolyte solvents, additives, and salt(s) such that the outermost active material portions passivate and do not further react with the electrolyte, which is particularly true for graphite anodes. Second, the liquid electrolyte must remain in intimate contact with the active materials throughout the formation process using a step referred to as "wetting," which may be completed at elevated temperature. Elevating the wetting temperature to 40°C–60°C reduces the liquid electrolyte contact angle and provides access of the electrolyte to the electrode mesopores. Without excellent electrolyte wetting of the electrode macropores and mesopores, complete SEI and CEI formation will not occur for a given protocol for large cells. For this reason, industry includes multiple wetting steps during formation cycling. Significant strides have also been made in the fundamental understanding of the physical properties of electrodes and electrolyte and wetting relationship.[3,4] Third, the charging and discharging steps are completed at slow rates to ensure the thickness and composition of the SEI and CEI are optimized for minimum capacity fade over the cycle life of the battery. A tap charge is also added after electrolyte injection to increase cell voltage and avoid Cu foil corrosion. Finally, the cost and amount of capital equipment to complete the formation of every cell produced is either prohibitive for ultra-high cell throughput or creates a production bottleneck for limited cell throughput.

The wetting and formation steps are followed by an aging step, which may take an additional 1–2 weeks to complete, where leak currents are checked. Leak currents range from about 20 to 50 $\mu A/cm^2$ immediately after formation for a few hours and taper off to ~2–5 $\mu A/cm^2$ after a couple of days, and finally to <1 $\mu A/cm^2$ after a couple of weeks. This aging process also requires many electrochemical cyclers, environmental chambers, and large amounts of associated floor space, up to one-quarter of the size of a typical battery plant. Cells may be grouped during a labor-intensive process called "binning" based on their leak currents and rated capacity after formation cycling, which is for reducing the cell-to-cell capacity variation. Reduction in either time or equipment would greatly benefit the production of LIB manufacturing.

There have been several excellent reviews written about the SEI formation process and its reaction mechanisms, as well as its characterization from a chemical, morphological, and materials standpoint;[5–8] however, this article will focus on the state-of-the-art processes used in the wetting and formation and the future outlook on how to make them faster, more cost-effective, and less capital intensive. In addition, an emphasis will be placed on the graphite anode, as its SEI formation process is much more intricate than other anode materials such as $Li_4Ti_5O_{12}$ (LTO), which do not form appreciable SEI layers. It is also important to note that other advanced batteries based on alkali-metal ion transport such as sodium-ion, Li-S, and all-solid-state batteries have their own unique SEI formation challenges with metal anodes and are beyond the scope of this article.

Little information has been published in the open scientific literature on LIB formation-cycle time reduction, but there have been several important studies over the past 5–10 years. An et al. was the first to publish a systematic study on the reduction of formation time and concluded that similar long-term cycling performance to a slow baseline could be obtained by simply increasing the charge/discharge rates and reducing the number of initial full cycles.[9] Placke et al. studied the oxygen-group surface chemistry and ratio of basal to edge planes in graphite anodes and their effect on irreversible first charge/discharge capacity. They found that the irreversible capacity loss after the first discharge was highly linked to the graphite edge-plane surface area and extent of bound polar groups.[10] This finding illustrates how the morphology and surface chemistry of the graphite is related to first-charge rate and the wetting steps of the formation protocol. Another study by Lee on the effect of fluoroethylene carbonate (FEC) addition to the electrolyte combined with shallow cycling showed that Coulombic efficiency was improved during SEI formation cycling by formation of inorganic nanoparticles on the anode graphite surfaces.[11] It was also found that long-term cycling performance improved as well, and this result is one of the few published ones that clearly shows the benefit of electrolyte additives.

Discussion

Dating back to the early 1990s, the wetting process in the LIB industry took many days and the charge/discharge formation cycling was completed in multiple cycles at rates as little as 0.05C/−0.05C. Over the last 25+ years,

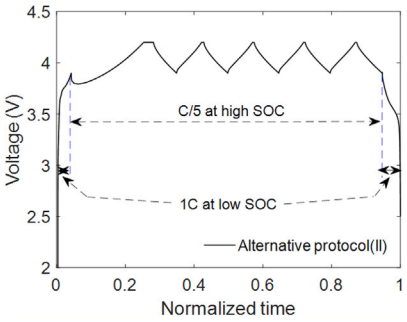


Figure 1. Ultra-Fast Formation Protocol of 14 h (8.5× Faster Than 3 Full 0.05C/−0.05C Cycles) Utilizing Fast First Charging, Shallow Top-of-Charge Cycling (Charging and Discharging), and Fast First Discharging

the SEI formation process has been improved, but it still takes 3–7 days to complete. A typical SEI formation process may involve a first electrode wetting for 6–24 h at room temperature, 1–2 formation cycles at 0.1C–0.2C charge/discharge rates, a second electrode wetting for 12–24 h at room temperature, 1–2 formation cycles at 0.2C–0.5C, and finally a third electrolyte wetting for 12–24 h at 40°C–60°C. This example is just one of many possible sequences, and actual SEI formation protocols are closely guarded industrial trade secrets, as well as the additives used to control and optimize SEI and CEI composition and structure and reduce the layer formation time.

Oak Ridge National Laboratory (ORNL) has developed ultra-fast formation protocols in 1.5 Ah full graphite/$LiNi_xMn_yCo_{1-x-y}O_2$ (NMC) pouch cells. A combination of high rates during the first charge and shallow cycling within a window of about 300 mV near the full state of charge (i.e., cell operating voltages of between about 3.9 V and 4.2 V) for the SEI and CEI formation have been successfully implemented. The shallow cycling is followed by a full fast discharge to complete the formation cycle, and the whole process is depicted graphically in Figure 1. When compared to a baseline protocol of three successive 0.05C/−0.05C cycles, the improved formation protocol is 8.5 times faster at only 14 total hours. This formation protocol was designed to take advantage of spending as much time in the low anode overpotential region as possible where the SEI forms via electrolyte solvent and salt decomposition. It was also developed for NMC 532 cathodes and could need further optimization for next-generation high-Ni cathode CEIs.

From an initial performance standpoint, the much faster formation cycling lowered the rated formation capacity as seen in Figure 2. For example, the fastest formation protocol (14 h) depicted in Figure 1 gave ∼146 mAh/g as compared to ∼168 mAh/g for the slow baseline formation, a 13% lower rating. When completing the first partial charge and first full discharge at slightly slower rates (0.2C and 0.33C as shown in Figure 2), the difference between the rated capacity of the baseline and the ultra-fast formation dropped to only ∼12 mAh/g, or 7% lower. It is well known that discharge capacity decreases slightly between the range of −0.05C and −1C, and the first discharge capacities in Figure 2 represent this performance difference.

These initial capacity differences may seem unacceptable from a manufacturer's viewpoint, and likely even result in different morphologies and compositions of the respective anode SEI layers, as found by Bhattacharya and Alpas when investigating the effects of different cyclic voltammetry rates on anode graphite.[12] They found via detailed high-resolution TEM imaging that at high scan rates of 3 mV/s a non-uniform, columnar SEI coverage of the anode graphite occurred, and at low scan rates a uniform, smooth, tubular coverage occurred. The morphology and degree of inorganic compounds formed in the SEI also varied with scan rate. However, these findings do not seem to make a difference from a long-term cycling standpoint. Stable SEI and CEI layers must still be forming during the shallow (near full state of charge) cycling even though not all the available capacity is accessed in just one full charge/discharge cycle. This hypothesis is evidenced by the long-term cycling data shown in Figure 3, where it is seen that the performance after 200–300 1C/−1C cycles is identical for the four datasets. It should be noted that the same basic electrolyte with no additives was used in all cases for the data shown in Figures 2 and 3. This electrolyte has a composition of 1.2 M $LiPF_6$ in a 3:7 weight ratio of ethylene carbonate (EC) to diethyl carbonate (DEC).

Outlook

Combined electrolyte wetting and SEI/CEI formation protocols continue to be a major expense and processing bottleneck for LIB manufacturers. Electrolyte penetration into the electrode pores (z direction) is as important as spreading over the surface (x-y direction) for

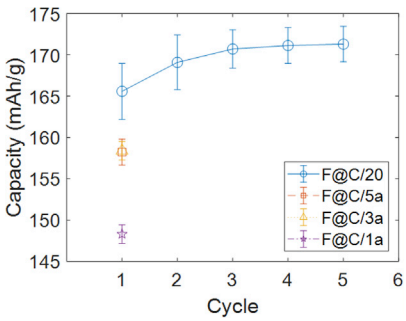


Figure 2. Discharge Capacities after Different Formation Protocols

Rated discharge capacities after ultra-fast formation cycling (according to the protocol shown in Figure 1) showing a drop in first-cycle capacity compared to the C/20 and −C/20 charge/discharge baseline (blue circles); C/5 and −C/5 charge/discharge cycling up to and below 3.9 V (red squares), C/3 and −C/3 charge/discharge cycling up to and below 3.9 V (orange triangles), C/1 and −C/1 charge/discharge cycling up to and below 3.9 V (purple stars). All cells evaluated were 1.5 Ah full graphite/LiNi0.5Mn0.3Co0.2O2 pouch cells in triplicate (error bars represent a 2σ standard deviation for each group of 3 cells). Note: the baseline protocol has five full discharges, whereas the improved shorter protocols only have one full discharge.

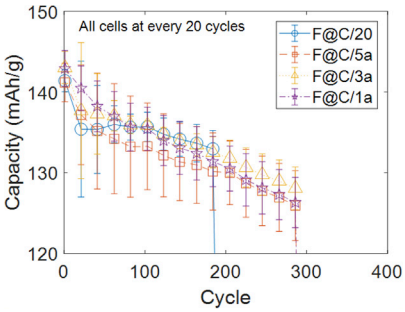


Figure 3. Capacity Retention Comparison after Different Formation Protocols

Long-term cycling at 1C/−1C after ultra-fast formation cycling (according to the protocol shown in Figure 1) as compared to the baseline (five cycles at C/20 and −C/20 charge/discharge, blue circles); C/5 and −C/5 charge/discharge formation cycling up to and below 3.9 V (red squares), C/3 and −C/3 charge/discharge formation cycling up to and below 3.9 V (orange triangles), C/1 and −C/1 charge/discharge formation cycling up to and below 3.9 V (purple stars). All cells were processed and assembled identically and evaluated in 1.5 Ah full graphite/LiNi0.5Mn0.3Co0.2O2 pouch cells in triplicate. Error bars represent the performance of each group after every 20 cycles.

minimizing both initial electrolyte uptake and microporous wetting during the formation protocol, and these wetting phenomena are key to developing other fast formation protocols for large-format cells. A combination of rapid first charging and discharging with intermittent shallow cycling at the top of charge is an effective strategy to at least partially build the SEI layer. The rest of the SEI/CEI layer can then be formed during the first few service cycles of the LIBs, providing equivalent cell life as if the entire SEI/CEI layer was completed during the formation protocol. These findings provide a critical development pathway for industry to substantially reduce cell production and plant capital costs.

It may be possible to eliminate the wetting, formation, and aging steps altogether, however, using such technologies as atomic layer deposition (ALD) coatings on the active material powders, applying so-called "artificial" SEI and CEI layers, or implementing active materials that do not require a complicated SEI formation (such as LTO or Li metal anodes). In fact, LTO does not form an anode SEI at all. This undertaking may be dependent on the application, though, i.e., whether the LIB is for portable electronics, transportation, load leveling, or frequency regulation. For example, ALD coatings on the anode and cathode active materials are a novel approach for reducing wetting and formation cycle time. We anticipate that the improvements made at ORNL could be expanded with the ALD coating approach with the possibility of reaching formation times of <10 h, which would be an approximately 50% further reduction. It is expected that the protective ALD coating could even act as an artificial anode SEI itself and reduce the time needed at low anode potentials to form the inorganic components of the SEI from electrolyte salt decomposition. In an analogous fashion, the cathode could be protected with an ALD coating, which would reduce the need for a thick CEI for preventing transition metal dissolution and gas formation from electrolyte solvent decomposition at high cathode potentials.

Separators with advanced coatings could be used with novel materials applied to the external geometric or internal pore surfaces to enhance electrolyte wetting to a spontaneous level, as the current electrolyte solvents wet the state-of-the-art polyolefin separators quite poorly. In addition, the aging time could be dramatically reduced when leak currents of the cells are determined after the formation cycling. There is some evidence that a separator with ceramic coated porosity and higher electrolyte wettability can reduce leak currents during high-voltage holds. Vadivel et al. showed that reducing electrode cross-talk, transition metal dissolution, and gas generation crossover from the cathode to the anode via a non-porous, glass separator was key to reducing leak current magnitude.[13] An ALD coated separator could reduce the amount of time needed to predict what the leak current is (i.e., not needing to wait for the full asymptotic decay period) as well as reduce the magnitude of leakage. With such advancements to the active material powders and separator, a significant reduction in cell processing costs and capital equipment savings (or significant increase in throughput with the same equipment) could be realized.

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DECLARATION OF INTERESTS

A U.S. patent application has been filed by Oak Ridge National Laboratory related to this work: D.L.W., J.L., and S.J.A., "Fast Formation Cycling for Rechargeable Batteries," filed December 19, 2018, U.S. Patent

Application No. 16/225,889 (UT-Battelle, LLC).

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